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1 Constructed Laterality versus Learned Movement Information: A 2 Source-Held-Out Controlled Evaluation of Self-Supervised Gait 3 Representations

4 Abstract

5 We separate a geometric constraint we supply from information the encoder actually learns during
6 pretraining, and report what each contributes to a movement representation. We run a source-held-
7 out controlled evaluation of a compact S-JEPA-inspired encoder trained on GAVD-derived 3D pose,
8 across five outer folds and five seeds on 625 sequences from 93 sources. Reflection augmentation re-
9 duces a strict token-symmetry error q by 0.00843 (95% source-bootstrap interval [0.00687, 0.01020])
10 on the magnitude of the reduction, roughly 7.4% relative to the vanilla variant), so it measurably im-
11 proves consistency under a specified anatomical mirror action. Whether that helps prediction stays
12 unresolved: the native learned encoder’s advantage over its matched initialization is -0.01798 in
13 source-balanced R^2 , with an interval that spans zero. To isolate what pretraining contributes, we
14 build an odd readout whose construction guarantees output sign reversal under reflection — and its
15 learned features underperform matched initialization by 0.05874 in R^2 , an interval lying entirely
16 below zero. A separate classification pilot shows the same ordering, with raw pose reaching 0.441
17 macro-F1 against 0.292 for learned features on 20 held-out sources. Our contribution is a bounded,
18 artifact-verified evaluation protocol for this setting; the results support no claim of a pretraining
19 advantage and no claim of clinical validity.

20 1. Introduction

21 Representations of body movement are increasingly proposed as a foundation for downstream behav-
22 ioral and clinical readouts, yet a single accuracy number rarely settles whether such a representation
23 is doing useful work. Two questions matter to a Brain-and-Body foundation-model audience and
24 answer to different evidence: whether the representation carries information a downstream readout
25 can use, and whether it behaves correctly under a change of anatomical coordinate, in particular a
26 left/right reflection of the body. A representation can satisfy one while failing the other, so we treat
27 them as two separate axes of quality.

28 Video-derived pose adds a nuisance that inflates apparent performance, because a single upload
29 often yields several excerpts of the same person under one camera and setting, and splitting them in-
30 dependently lets closely related material land on both sides of a train/test divide so a readout scores
31 well by recognizing the recording. We therefore evaluate with source-held-out splits throughout en-
32 coder fitting, readout selection, and testing, following the standard grouped-cross-validation control
33 against shared-upload dependence [20]. This is weaker than held-out-person evaluation, because the
34 dataset supplies no persistent identity and the same person may recur across uploads.

35 Our setup is a compact skeleton joint-embedding predictive architecture inspired by S-JEPA [1] and
36 the latent-prediction principle of I-JEPA [2], trained on pose derived from the Gait Abnormality
37 in Video Dataset [19]. It groups 33 landmarks over time into short patches and predicts masked
38 latent content against an exponential-moving-average target encoder, given no laterality label and

no diagnosis, only pose geometry. Against this backdrop the laterality study asks whether learned features predict a signed left/right motion contrast whose sign is defined to reverse under anatomical reflection, whether reflection augmentation during pretraining improves how the representation transforms under that reflection, and what an imposed sign rule contributes once added to the readout. The signed target and the reflection operation are constructed analytically, the readout weights are fitted from pose-derived values, and the odd readout’s sign law — the guarantee that a mirrored input flips the output — is imposed by construction, so it holds for any encoder and any fitted weights. This keeps the learned, the fitted, and the built-in components distinct throughout.

Contributions. None of the individual ingredients is new: grouped cross-validation, linear probing, parity projection, and building geometric structure into learning are all established [20, 7, 6], and our contribution is their controlled combination as a test. We give a two-axis evaluation for movement representations, downstream predictive information and behavior under anatomical reflection, and show that the two can diverge, so correct sign behavior does not imply predictive utility. We define a signed left/right motion contrast with an exact reflection sign law and separate three claims that are easily conflated: analytic output oddness imposed by the readout, probe-level sign behavior, and intrinsic token equivariance of the encoder. We run this under a leakage-controlled, source-held-out protocol with a paired untrained-encoder floor, five outer folds, five seeds, and source-cluster bootstrap uncertainty, which isolates what pretraining adds beyond initialization and supplied anatomy. A classification pilot runs as a secondary thread, comparing learned features against raw-pose and missingness baselines. The findings are bounded: reflection augmentation measurably improves the token-symmetry metric, its predictive gain is unresolved against the initialization floor, and an odd readout guarantees output sign reversal while its learned features underperform matched initial features. We claim no pretraining advantage, no clinical validity, and no foundation-scale result.

2. Background and related work

Our study draws on four literatures rather than extending any one of them. Self-supervised models for articulated motion descend from latent-prediction objectives: I-JEPA predicts the representation of masked image regions in an embedding space rather than reconstructing pixels [2], and S-JEPA carries that idea to skeleton sequences [1], as do related schemes such as Skeleton2vec [24] and masked motion predictors [15], while supervised graph models like ST-GCN [25] remain a standard reference for skeleton action recognition. Because latent prediction can collapse to constant features, we regularize with the variance, invariance, and covariance terms of VICReg [3]. Our encoder is a small independent reimplement of the S-JEPA design trained on our own cohort, so we neither reproduce nor rely on any published benchmark number.

How symmetry enters a model is a second thread. Group-equivariant networks build a transformation group into the weight-sharing structure [7], the geometric deep learning program frames architectures through the symmetries they respect [6], and $E(n)$ -equivariant graph networks give a concrete instance for skeletal data [23]. When equivariance is not wired into the layers it can be obtained at the input or readout through data augmentation, frame averaging over the group [18], or a learned canonicalization map [14], ranging from letting the objective discover the symmetry to guaranteeing it by construction. The two mechanisms we study, reflection augmentation and an odd-parity readout, are ordinary points on that spectrum; we introduce no new equivariance algorithm and only measure where a self-supervised skeleton encoder falls on it.

The evaluation target itself comes from the clinical-biomechanics literature on gait symmetry, where signed left-versus-right contrasts expressed as symmetry indices are a long-standing summary of how the two sides of the body move [21, 22, 16, 4]. That work justifies treating a normalized left-minus-right motion contrast as a movement quantity worth probing, though our own quantity is a dimensionless index of estimated landmark speeds, chosen so that it provably negates under reflection, and not validated as a clinical asymmetry measure. Our inputs come from marker-free pose estimation, which has made video gait analysis practical without laboratory instrumentation [13], using the 33-landmark BlazePose schema of an on-device 3D pose estimator [12, 11]. Self-supervised and forecasting objectives have been applied to pose-based gait for movement-disorder severity [9] and fine-grained abnormality detection [8]; those studies ask whether a learned representation predicts a clinical outcome, whereas we ask only whether it carries a specific bilateral-symmetry structure that behaves correctly when the input is mirrored.

94 3. Methods

95 **Cohort and unit of evaluation.** Both studies draw on the Gait Abnormality in Video Dataset
 96 [19, 10], whose distribution supplies annotations and public video links rather than raw media, with
 97 folder labels (normal, Parkinson’s, stroke, myopathic, cerebral palsy) that this project has not vali-
 98 dated as diagnoses. A single upload can yield several sequences, so the *source* is a coarser unit than
 99 the *sequence*, and splitting sequences independently would let clips from one recording fall on both
 100 sides of a train/test divide; every fold is therefore built at the source level. The laterality study (Study
 101 B) is our primary evidence, with pre-specified quality and target-computability gates retaining 625
 102 sequences from 93 sources out of 642 available derived-pose archives; the classification pilot (Study
 103 A) is a separate, earlier design on an overlapping cohort of 639 sequences from 97 sources, so the
 104 two are distinct designs whose counts must not be pooled. Source grouping controls shared-upload
 105 dependence [20], but GAVD supplies no persistent person identifier and we do not infer one, so
 106 source-disjoint evaluation is a more limited guarantee than person-disjoint evaluation. Appendix A
 107 gives the per-class composition and full attrition ledger.

108 **Representation learning.** Both models share a skeleton front end that groups 33 pelvis-centered,
 109 scale-normalized BlazePose landmarks into four-frame patches over a 64-frame window and pre-
 110 dictors masked latent content against an exponential-moving-average target encoder, with neither the
 111 laterality target nor the annotation label entering the objective. Study B uses a 96-dimensional four-
 112 layer encoder with a two-layer Transformer predictor, trained with a centered latent cross-entropy
 113 regularized in a two-view VICReg style [3]. The one comparison that matters here is between a
 114 vanilla variant that never reflects a training sample and a reflection-augmented variant that reflects
 115 each sample with probability 0.5 before its two views are generated; within a seed the two variants
 116 share initialization, source draws, masks, views, and optimizer schedule, so the pair stays matched.
 117 Appendix A gives the full architecture, loss weights, and schedule.

118 **A signed laterality target.** The evaluation target compares movement on the two sides of the
 119 pose. Let M negate the horizontal coordinate and swap the full anatomical left/right landmark map,
 120 including validity flags; applied twice it returns the input, so $M^2 = I$. Five landmark pairs enter the
 121 target (shoulders, knees, ankles, heels, foot tips), with the hips excluded because pelvis-centering
 122 makes their two velocity magnitudes algebraically identical. For each pair k we take the median
 123 left and right speeds $m_{L,k}$ and $m_{R,k}$ over paired-valid frame transitions and form the normalized
 124 contrast and target

$$c_k = \frac{m_{L,k} - m_{R,k}}{m_{L,k} + m_{R,k} + 10^{-8}}, \quad y(X) = \frac{1}{5} \sum_{k=1}^5 c_k.$$

125 Because M swaps every left landmark with its right partner, $y(MX) = -y(X)$ holds even for an
 126 asymmetric observed gait, which is the property this evaluation relies on. Across the cohort the target
 127 is small and roughly centered (mean -0.00607 , standard deviation 0.05915). It is a dimensionless
 128 coordinate-derived index, not a validated clinical asymmetry measure or a marker of neural laterality
 129 [16].

130 **Separating learned prediction from the imposed sign rule.** For each of the five pairs an anatomi-
 131 cal feature $A(X)$ concatenates the left-minus-right and left-plus-right token means on the shared
 132 valid temporal support, giving 960 features supplied equally to the trained encoder and to its
 133 recorded initialization, so any advantage of training must appear on top of the same structure. From
 134 A we form equal-width odd and even projections under reflection,

$$\Phi^-(X) = \frac{A(X) - A(MX)}{\sqrt{2}}, \quad \Phi^+(X) = \frac{A(X) + A(MX)}{\sqrt{2}},$$

135 and since $M^2 = I$ the odd part satisfies $\Phi^-(MX) = -\Phi^-(X)$. The constructed odd readout
 136 is $h(X) = w^\top \Phi^-(X)$ with feature centering and the regression intercept both disabled, since a
 137 nonzero intercept or feature-mean offset would break the identity $h(MX) = -h(X)$ that otherwise
 138 holds for any encoder and any fitted w . The weights still learn to predict y , but the sign behavior
 139 follows from the construction alone and reveals nothing about acquired information [7, 18]. We
 140 evaluate four readout lanes — native, odd, odd with zero origin, and even — each fitted for both the

141 learned features and their matched initialization, with the ridge penalty selected on four inner source
142 folds within the outer-training set.

143 **Strict token-symmetry metric.** A separate test asks whether the encoder tokens themselves trans-
144 form correctly under reflection. We compare the tokens of a mirrored input against the anatomically
145 permuted tokens of the original, allowing no Procrustes alignment, rotation, sign search, or fitted
146 transform, and report a normalized squared error q on the jointly valid positions (Appendix A gives
147 the exact form). A value near zero indicates equivariance under this action, while unrelated equal-
148 energy representations sit near one. The metric is uncentered, so features that are shared across
149 inputs or insensitive to the input can lower q without encoding anatomy, and a small value at initial-
150 ization does not by itself demonstrate learned structure; we treat 0.10 as an operational, uncalibrated
151 threshold.

152 **Estimand and uncertainty.** Study B divides the 93 eligible sources into five stratified outer folds,
153 each holding out 18 or 19 test sources while the rest train the fold-local encoder and fit the readout,
154 so each source is a test source once and clips are weighted inversely to their source’s clip count.
155 The primary estimand is specific: for each seed we pool the held-out clip predictions across all
156 five outer folds, compute the source-balanced sequence-level R^2 , then average across the five seeds.
157 Uncertainty comes from 2,000 paired bootstrap draws that resample the 93 source clusters, keeping
158 each sampled source’s clips and all five seeds together. The resulting pointwise 95% intervals are
159 conditional on the fitted models and the fixed split; they do not propagate uncertainty from full
160 retraining or from selection over splits [5, 17].

161 4. Results

162 We report the laterality study first because it carries the controlled comparisons, and close with the
163 classification pilot as supporting context. Table 1 collects every contrast with its pointwise 95%
164 source-bootstrap interval, and the figure in Appendix A shows the same values graphically; the text
165 below reads the table rather than repeating it.

166 **Predictive value of the learned representation.** The native learned readout on the vanilla encoder
167 reaches an absolute R^2 of 0.05979, but its interval crosses zero, so on its own it does not clear a
168 floor of useful prediction. The paired contrast against the encoder’s own untrained initialization is
169 more informative, because it isolates what pretraining adds while holding fixed the variance shared
170 between readout and target. That difference is -0.01798 with an interval that spans zero, so the
171 predictive gain from this pretraining recipe is unestablished: the data neither show that pretraining
172 helped nor, given the interval width, establish equivalence to the untrained representation. Reflection
173 augmentation leaves the predictive estimates essentially unchanged, its augmented-minus-vanilla
174 contrast also crossing zero.

175 **Reflection augmentation and token symmetry.** The strict token metric gives a more definite read-
176 ing on the geometry. Augmentation measurably reduces the token-symmetry error relative to vanilla,
177 by -0.00843 with an interval entirely below zero, a reduction of roughly 7.4%, and it is the one con-
178 trast we report whose interval lies wholly on one side of zero. Two cautions attach to it. Both trained
179 variants have higher q than the initialization ($+0.03055$ and $+0.02213$), so training as configured
180 moves the tokens away from the approximate reflection consistency present at initialization and aug-
181 mentation only partly offsets that. And both trained variants have absolute q intervals that cross
182 the operational threshold of 0.10, so neither sits wholly below it, and this one fixed swap does not
183 exclude other latent symmetry actions.

184 **An imposed sign rule does not confer predictive benefit.** Because feature centering and the inter-
185 cept are disabled and the readout acts on the odd component Φ^- , the sign law $h(MX) = -h(X)$
186 holds for any encoder and any fitted weights, and the numerical check confirms it: for every seed
187 the original and mirrored predictions sum to zero within 10^{-6} . That guaranteed antisymmetry does
188 not translate into predictive value from pretraining. The constructed odd/zero-origin lane on the
189 learned vanilla encoder underperforms its matched initialization by -0.05874 with an interval lying
190 entirely below zero, so the modest structure these lanes exploit comes from the parity wrapper on
191 the anatomical pairs and the learned features add nothing.

Table 1: Paired laterality contrasts (Study B). ΔR^2 favors the first-named condition when positive; Δq favors it when negative. Intervals are point-wise 95% source-bootstrap intervals conditional on the fitted models and the fixed split.

Contrast	Metric	Estimate	95% interval
Native learned vanilla (absolute)	R^2	0.05979	$[-0.02527, 0.12571]$
Native learned – paired init.	ΔR^2	−0.01798	$[-0.03851, 0.00248]$
Native augmented – vanilla	ΔR^2	+0.00408	$[-0.00556, 0.01277]$
Constructed odd/zero learned vanilla (absolute)	R^2	0.04302	$[-0.04356, 0.11283]$
Constructed odd/zero learned – matched init.	ΔR^2	−0.05874	$[-0.09549, −0.01740]$
Initialization (absolute)	q	0.08322	$[0.07458, 0.09404]$
Vanilla (absolute)	q	0.11377	$[0.09513, 0.13843]$
Augmented (absolute)	q	0.10534	$[0.08783, 0.12778]$
Vanilla – initialization	Δq	+0.03055	$[0.01576, 0.04763]$
Augmented – initialization	Δq	+0.02213	$[0.00844, 0.03934]$
Augmented – vanilla	Δq	−0.00843	$[-0.01020, −0.00687]$

192 **Classification pilot (supporting context).** The earlier five-class pilot in Study A offers context
193 but not confirmation. On the 20 held-out sources of its one retained execution (outer fold 0, seed
194 42), a classifier built on raw pose features reaches a test macro-F1 of 0.441, above the learned-
195 representation lane at 0.292 and the missingness-only baseline at 0.251, yet all three lanes miss
196 every stroke source in the split. Because this is a single fold and seed with no matched untrained-
197 encoder control and an overlapping cohort, the controlled comparisons above neither explain nor are
198 corroborated by it. Appendix B gives the per-lane scores and the full configuration.

199 5. Discussion

200 The main result is simple to state. In Study B we produce a physically meaningful output whose
201 sign flips under a left/right reflection of the body, as a laterality index should, without that behavior
202 depending on the features produced during pretraining: the odd readout guarantees $h(MX) =$
203 $-h(X)$ for any encoder and fitted weights, and reflection augmentation measurably improves the
204 strict token-symmetry metric, so the correct reflection behavior comes from the group-theoretic
205 projection and the data augmentation we added rather than from what the encoder learned. Against
206 that, the pretrained representation contributes no demonstrated predictive value over its own starting
207 point, with the native contrast straddling zero and the sharper odd/zero-origin lane entirely negative;
208 both trained variants also carry higher q than the initialization, so on the one symmetry action we
209 specified training shifted the representation away from consistency even as augmentation partly
210 counteracted that.

211 Read against this workshop’s themes of movement modeling and pretraining evaluation, these find-
212 ings argue that a correctly-transforming, interpretable output is a weak signal about the content of
213 a body representation. A reviewer who saw only the odd readout’s clean sign reversal, or only
214 the augmentation-driven improvement in q , might credit the encoder’s features, and the matched-
215 initialization controls show that credit would be misplaced. Evaluating pretraining for movement
216 data therefore needs three things measured side by side: downstream accuracy, a difference against a
217 matched untrained initialization on the same probe and split, and explicit transformation-law checks
218 under a specified group action. The classification pilot points to the same caution, since there a sim-
219 pler raw-pose baseline outscored the pretrained features. Our methods — grouped source-level split-
220 ting, linear probing, latent masked prediction, and symmetry projection through group-equivariant
221 and frame-averaging machinery — are established rather than new, and the contribution is the con-
222 trolled way we combine them on gait data, which we offer as a robust reference point for anyone
223 asking whether pretraining helps a body model.

6. Limitations and responsible use

The evidence here is bounded. The two studies draw on overlapping GAVD-derived cohorts, the same individual may recur across uploads, and we have no reliable person-level grouping, so they are not independent replications, must not be pooled, and can control leakage only at the source level; with no external cohort, nothing here speaks to generalization outside GAVD. The Study B target is a dimensionless index engineered to satisfy $y(MX) = -y(X)$, not a clinically validated measure of gait asymmetry, proprioception, or neural laterality. The strict token metric q is uncentered and probes a single anatomical-swap action, so shared or input-insensitive features can lower it, a low value does not by itself prove learned anatomical structure, other symmetry actions are not excluded, and the 0.10 figure carries no population-level meaning. The linear readouts are small-sample, so a null probe result bounds what these probes detect, not what the encoder contains. Study A rests on a single fold and seed with no matched untrained-encoder control, and its checkpoint and evaluation bundles are absent from the checkout, so its hyperparameters follow code defaults and Study B’s controls do not retroactively repair it. None of this is foundation-scale, and we present no neural-mechanism, multimodal-transfer, or closed-loop control evidence.

Responsible use of the underlying data carries its own constraints. GAVD aggregates publicly posted gait videos [19, 10], but public availability does not by itself establish informed consent or permission to reuse, and gait trajectories can be identifying even after pose extraction strips the pixels. The project’s ethics determination, data-use review, and derived-pose release review are all unresolved, and our own reporting sets the submission and release gate to closed; a computed readiness index cannot override that gate. We make no clinical, surveillance, or deployment claim, and nothing here should be read as licensing these methods or this derived data for screening or diagnosis of any person.

7. Conclusion

Our controlled evaluation shows that a physically interpretable output can transform correctly under body reflection by construction while the pretrained representation shows no predictive advantage over its matched initialization on these probes. The finding is bounded to overlapping observational cohorts, a clinically unvalidated coordinate-derived target, and a single specified symmetry action. Extending it would take an independent cohort with reliable person grouping, a validated laterality target, centered and alternative symmetry actions, and a laterality-supervised objective matched to the same evaluation, none of which we carry out here.

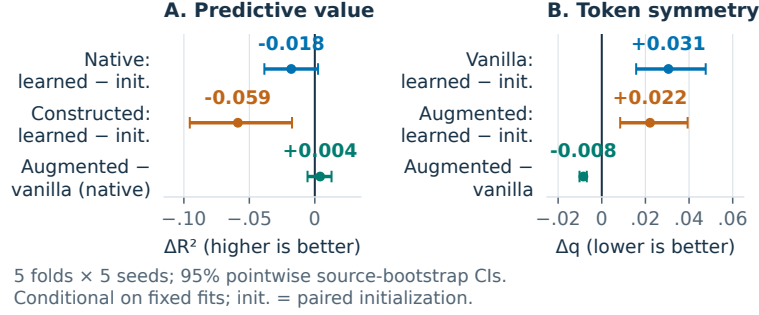


Figure 1: Paired laterality contrasts across five source folds and five seeds, with pointwise source-bootstrap intervals, plotting the same values as Table 1. Panel A shows predictive R^2 differences (larger favors the trained representation); Panel B shows the strict token-symmetry error (smaller favors it).

Appendix A. Study B implementation and reproducible detail

Cohort composition and attrition. The frozen GAVD inventory describes 666 annotated sequences spanning 103 source uploads and 140,641 annotated frames. Starting from 642 available derived-pose archives, pre-specified quality and target-computability gates retain 625 sequences from 93 sources and exclude 17. The retained cohort holds 270 normal, 39 Parkinson’s, 75 stroke, 183 myopathic, and 58 cerebral-palsy sequences, drawn respectively from 29, 9, 18, 28, and 9 sources. Source roles are frozen before downstream attrition removes any sequence.

Skeleton front end. Each pose sequence carries 33 BlazePose landmarks (MediaPipe schema) with horizontal, vertical, and depth coordinates; the pose is centered on the pelvis and scale-normalized, and time is grouped into four-frame patches over a 64-frame window. Twelve anatomically named landmarks are eligible masked targets: the shoulders, hips, knees, ankles, heels, and foot tips (MediaPipe indices 11, 12, and 23 through 32). Each four-frame patch flattens its four XYZ observations rather than averaging them, giving the encoder access to within-patch motion, and patches with no valid content are padded out of attention.

Encoder, predictor, and objective. The encoder is 96-dimensional with four layers and four attention heads, followed by a two-layer Transformer predictor with four heads. During training up to 60% of the eligible valid target positions are masked, limited by the batch minimum, and the model predicts the hidden latent content against an EMA teacher. The objective is a centered latent cross-entropy with student and teacher temperatures 0.10 and 0.06, regularized in a two-view VICReg style [3]:

$$L_B = L_{\text{latent CE}} + 0.05(25 L_{\text{invariance}} + 25 L_{\text{variance}} + L_{\text{covariance}}).$$

Optimization and augmentation. Each of 300 epochs draws one clip per training source and then pads to a source-uniform four batches of 20, producing 1,200 gradient updates per encoder under AdamW at learning rate 10^{-3} , betas (0.9, 0.95), weight decay 0.05, cosine decay, and gradient-norm clipping at 1; the EMA coefficient rises from 0.999 toward 1 with target-centering coefficient 0.9. The two views apply rotations up to eight degrees and XY translations up to 0.03 normalized units. The vanilla variant never reflects a training sample; the reflection-augmented variant reflects each sample with probability 0.5 before the two views are generated. Within a seed the two variants share initialization, source draws, masks, views, optimizer schedule, and update count. Checkpoints follow a fixed schedule with no early stopping and no validation-based selection.

Target computation. The five contributing pairs are shoulders, knees, ankles, heels, and foot tips. For each pair the median left and right speeds are taken over paired-valid transitions, meaning both landmarks are observed at both ends of the same frame transition, with at least eight such transitions required per pair; speeds use the original timestamp differences and uninterpolated coordinates.

Strict token-symmetry metric. Let $Z(X)$ and $Z(MX)$ be target-encoder tokens and S the anatomical landmark permutation on the latent channels; on the positions C valid in both aligned views, with no Procrustes alignment, rotation, sign search, or fitted transform allowed, the strict error is

$$q(X) = \frac{\|Z(MX) - S Z(X)\|_C^2}{\|Z(MX)\|_C^2 + \|S Z(X)\|_C^2}.$$

A value near zero indicates equivariance under this action, while unrelated equal-energy representations sit near one. The metric requires at least eight jointly valid positions and rejects negligible-energy representations. It is uncentered, so features shared across inputs or insensitive to the input can lower q without encoding anatomy, and a small value at initialization does not by itself demonstrate learned structure; we treat 0.10 as an operational, uncalibrated threshold, and a different joint permutation could yield a different number.

Fitting and inner selection. Within each outer-training set, four inner source folds select the ridge penalty from $\{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100, 1000\}$ on outer-training embeddings only; these inner folds tune the readout and never stop encoder training or select a checkpoint. Source-level sign accuracy and balanced accuracy [5] are reported as secondary measures, and the fitting and scoring stack is the standard ridge and cross-validation machinery [17]. We report the source-balanced, seed-averaged sequence-level R^2 rather than a source-mean-target R^2 , which would score one aggregated prediction per source, or a seed-ensemble score, which would average predictions before scoring; each answers a different question.

Appendix B. Study A classification pilot

The pilot model is deliberately smaller and organized differently from Study B. Its four-frame patches are coordinate-averaged and read by a pooled-context MLP predictor, with the implemented loss

$$L_A = L_{\text{SmoothL1}} + 0.10 L_{\text{variance}} + 0.01 L_{\text{covariance}},$$

where an optional condition cross-entropy term (weight 0.10) is disabled in the primary run, though the annotation labels still set a cumulative normal-first exposure order: normal, then adding Parkinson’s, stroke, myopathic, and cerebral-palsy sequences, with earlier categories retained. The encoder is width 64 with two layers and four heads, and the recorded defaults are batch size 32, 20 epochs of 100 steps per stage, learning rate 10^{-3} , and EMA 0.996. Its quality checks retain 657 sequences from 100 sources with public metadata, 655 from 98 after media-span checks, and 639 from 97 after pose quality control.

Study A rests on a single retained execution (outer fold 0, seed 42), and its current checkpoint and evaluation bundles are absent from the checkout, so these hyperparameters follow code defaults, without an independently confirmed run configuration or a matched untrained-encoder control. On the 20 held-out sources of that fold, the three lanes score as follows.

Table 2: Classification pilot (Study A), 20 held-out sources of fold 0, seed 42. Feature dimensions are at the default width. This single execution has no matched untrained-encoder control, and all three lanes miss every stroke source in the test split.

Lane	Feature dim	Test macro-F1	Balanced accuracy
Raw pose	144	0.441	0.443
Learned representation	256	0.292	0.257
Missingness only	97	0.251	0.248

Appendix C. Evidence boundaries and release status

The table below records what each headline claim rests on and where its evidentiary boundary lies. It is meant to keep the reported findings inside the limits the data support.

Table 3: Claim, supporting evidence, and evidentiary boundary.

Claim	Evidence	Boundary
Odd readout guarantees output sign reversal Reflection augmentation lowers q vs. vanilla	Analytic (Φ^- with zero origin) plus per-seed numerical check to 10^{-6} $\Delta q = -0.00843$, interval $[-0.01020, -0.00687]$ below zero	Property of the construction, not of learned features One fixed anatomical swap; both trained variants still exceed the 0.10 threshold and the initialization
Pretraining predictive gain (native)	$\Delta R^2 = -0.01798$, interval $[-0.03851, 0.00248]$	Interval spans zero: neither a gain nor equivalence is established
Pretraining predictive gain (odd/zero)	$\Delta R^2 = -0.05874$, interval $[-0.09549, -0.01740]$ below zero	Trained features underperform matched initialization on the sign-guaranteed lane
Classification pilot ordering	Raw pose 0.441 vs. learned 0.292 macro-F1	Single fold and seed, no untrained control, overlapping cohort

323 The project’s ethics determination, data-use review, and derived-pose release review are all unre-
324 solved. Our own reporting sets the submission and release gate to closed, and a computed readiness
325 index cannot override that gate. We do not claim any approval that has not been granted.

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